

(TR-104) PYTHON WITH AI/ML

Training Day 20 Report

Date: 6 February 2026

Topic Covered: Logging, Exception Handling & Automation Scripts in Python

The twentieth day of training focused on understanding logging mechanisms, exception handling concepts, and automation script development in Python. The day included building real-world utilities such as log analysis tools and data processing pipelines.

- Understood why print() is not suitable for production debugging and learned logging as an alternative.
- Studied logging levels and proper log formatting techniques.
- Learned file rotation concepts for managing log files efficiently.
- Developed a log analyzer to read logs and report errors.
- Implemented centralized logging configuration for better monitoring.
- Handled logging of errors and performance-related information.
- Practiced exception handling including custom exceptions and global exception handling.
- Implemented retry mechanisms for API calls and developed fail-safe scripts.
- Built automation scripts including file cleanup automation.
- Created a CSV to JSON transformer for data conversion.
- Designed an API data pipeline for structured data handling.
- Prepared a cron-ready script for scheduled automation tasks.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/Python-Scripts>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/getting-started-with-python-scripts-a-practical-approach-49e656bd5862>